

MITIGATING SYNCHRONIZATION EFFECTS IN PRICE-AWARE EV CHARGING USING EQUITABLE, COMMUNICATION-FREE V-P CONTROL

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Abstract

Electric vehicle (EV) charging synchronization causes voltage violations in low-voltage (LV) distribution grids. To resolve these operational limits, this paper proposes a communication-free, decentralized adaptive voltage-power (V-P) charging strategy that combines grid stability, spatial fairness, and end-user economic optimization. We validate the proposed architecture through Monte Carlo simulations on the IEEE LV test feeder, integrating empirical residential EV charging data from France to capture stochastic real-world demand. We benchmark the algorithm against uncontrolled demand, traditional voltage droop, and price-optimized charging. Results demonstrate that the adaptive V-P controller matches the EN 50160 compliance of traditional droop control (0.0% of simulation runs resulting in voltage violations) while significantly improving spatial equity across the feeder. Economically, the strategy achieves the same average charging price (0.11 €/kWh) as price-optimized scheduling, outperforming both droop and uncontrolled baselines. Ultimately, the proposed controller increases the hosting capacity of the LV network while ensuring equitable curtailment distribution and minimizing end-user charging costs.

1 Introduction

Electric vehicles (EVs) are becoming increasingly common in the grid. This transition is important for mitigating local air pollution, particularly in residential areas, and reducing global greenhouse gas emissions, especially when EV charging couples with renewable energy sources (RESs). As for the connection location, recent analyses [1] indicate that most drivers connect their EVs to the low-voltage (LV) distribution grid via private residential or curbside AC chargers. However, most LV networks were not designed to host these high-energy, high-power loads, especially when they are synchronized [2]. This could cause severe operational stress, accelerating grid asset degradation, and causing voltage violations.

Synchronization of EVs can be driven by different mechanisms. The most immediate cause is natural arrival clustering, where users connect their EVs following standard commuter schedules. Then, financial incentives can generate a second form of synchronization. If users delay charging to exploit off-peak grid tariffs, they artificially concentrate demand into narrow and highly congested time windows. Finally, aggregators have started to provide transmission system operators (TSOs) services by utilizing the flexibility potential of EV fleets [3]. When these fleets respond to the same TSO power modulation signals, their aggregate reaction can generate instability and violations in the LV grid.

Researchers have explored various distributed and decentralized mitigation strategies to manage these integration problems, while reducing the risk of single-point failure exhibited

by centralized control methods [4, 5]. One of the simplest approaches is the standard voltage droop control, which aims to reduce grid violations by linearly decreasing charging power when voltage goes below a certain threshold. Yet, this approach inherently penalizes users who are connected far from the LV transformer and systematically experience a lower voltage. To correct this spatial unfairness, [6] introduces an equitable charging strategy that adapts power limits based on local voltage and voltage-to-load sensitivity. Their method decreases allowed power more aggressively at electrically stronger buses. However, control parameters need to be tuned after extensive system testing and therefore this strategy cannot be considered plug-and-play. Hossain et al. simplified this architecture by replacing the computationally expensive exponential function with a simpler linear formulation [7, 8]. Despite the higher efficiency, they leave two control parameters not clearly defined. In addition, both strategies incorporate state of charge into their algorithms, even though this metric is commonly not available to commercial AC chargers. Moreover, in both cases the charging strategy is tested either on small LV grid models, or on medium voltage (MV) grid models, with a limited focus on the LV part. Other researchers propose hardware-dependent or communication-heavy alternatives. For instance, [9] designed an equitable charging strategy requiring bidirectional chargers to inject reactive power through a piecewise function. The kind of hardware capable of performing such operations, though, is not commercially available yet. Similarly, [10] utilized individual voltage-to-load sensitivities to calculate a dynamic operating envelope. However, this method

requires continuous and bidirectional communication between the individual EVs and the LV transformer.

Collectively, these existing strategies present significant barriers to realistic commercial deployment. They lack true plug-and-play capability, since they require prior topological knowledge, machine learning training phases, or active communication frameworks between the charger and the grid. Moreover, methodologically most studies restrict their analysis to deterministic 24-hour horizons, failing to capture the stochastic variability inherent in multi-day empirical charging profiles [11]. Finally, while some sources [12] test their charging strategy in the scenario of Time-of-Use (ToU) tariffs, they rarely integrate user cost savings as an active control objective.

To bridge the gaps identified in the literature, this paper introduces a communication-free, plug-and-play adaptive voltage-power (V-P) control strategy. We propose three primary contributions.

- We develop a localized algorithm that integrates ToU and day-ahead price optimization directly into the droop logic.
- We refine existing decentralized impedance estimation methods to account for practical data exchange limits, enhancing their applicability in a real grid.
- We validate the proposed architecture through multi-day Monte Carlo simulations on the IEEE LV test feeder, fully incorporating stochastic user behavior.

The remainder of this paper is structured as follows. Section 2 details the proposed adaptive V-P charging strategy, combining equitable voltage droop with price-optimized scheduling. Section 3 defines the Monte Carlo simulation framework. This includes the invariant, parametric, and stochastic variables, alongside the benchmark strategies, operational scenarios, and key performance indicators. Section 4 presents the simulation results through a global performance assessment and a detailed specific scenario. Finally, Section 5 draws the conclusions.

2 Methodology

We designed the proposed charging strategy to achieve three objectives: practical implementation via standard data exchange, equitable power curtailment, and guaranteed economic savings. To satisfy these requirements, the control architecture synthesizes a fair voltage droop mechanism with a price-responsive scheduling algorithm into a single plug-and-play framework.

2.1 Equitable Voltage Droop and Impedance Estimation

To ensure operational fairness, each EV charger estimates its local voltage-to-load sensitivity. LV distribution grids are primarily resistive, making local voltage magnitude highly dependent on active power variations. Chargers calculate this sensitivity by measuring the local voltage deviation (ΔV) following a known change in active charging power (ΔP) [6]. Because sensitivity is inversely proportional to grid voltage (it is higher for lower voltages) comparing sensitivities directly would be unfair. To correct this, each charger records its

sensitivity under various loading conditions and extrapolates the value to the baseline voltage of 0.9 p.u. This extrapolation provides a standardized sensitivity metric that reflects the maximum expected grid stress.

The proposed strategy relies on a lightweight, asynchronous data exchange architecture. Once a charger calculates its standardized sensitivity, it transmits this single scalar value to the central aggregator. The aggregator collects these metrics from all participating chargers and maps them to a normalized sensitivity parameter σ_i , scaled between 0 and 1. The electrically furthest charger (with the highest standardized sensitivity) receives $\sigma_i = 1$, while the charger closest to the transformer (with the lowest standardized sensitivity) receives $\sigma_i = 0$. The aggregator broadcasts these normalized parameters back to the chargers at low frequencies, such as during initial connection or with daily updates. This architecture eliminates the need for continuous, real-time communication between chargers or other grid elements.

The local charger inputs the normalized sensitivity σ_i to its local voltage droop controller to adapt the power throttling threshold. Let V_i represent the measured local voltage at node i . The droop controller limits the active charging power, $P_{droop,i}$, according to a piecewise linear function. To respect the IEC 61851 standard, the controller sets the lower throttling power limit to 1.4 kW per phase. All chargers completely stop charging if the local voltage falls below the limit of 0.9 p.u. The upper voltage threshold, $V_{th,i}$, where power curtailment begins, adapts dynamically based on the sensitivity parameter. We define this adaptive threshold as $V_{th,i} = 0.99 - 0.08\sigma_i$. Consequently, a charger near the transformer ($\sigma_i = 0$) begins reducing power preemptively at 0.99 p.u. Instead, a distant charger ($\sigma_i = 1$) maintains maximum charging power, P_{max} , until the voltage drops to 0.91 p.u. We express the complete equitable droop function as follows:

$$P_{droop,i}(V_i) = \begin{cases} P_{max} & \text{if } V_i \geq V_{th,i} \\ \frac{P_{max}-1.4}{V_{th,i}-0.9}(V_i - 0.9) + 1.4 & \text{if } 0.9 < V_i < V_{th,i} \\ 0 & \text{if } V_i \leq 0.9 \end{cases} \quad (1)$$

This formulation ensures that power curtailment is distributed more evenly across the network.

2.2 Price-Responsive Scheduling

To minimize charging costs, the local controller processes a combined price signal, $C(t)$. We define $C(t)$ as the sum of the static seasonal distribution (ToU) tariffs and dynamic day-ahead energy prices. While ToU tariffs remain constant across a season, the energy market publishes day-ahead prices daily at 13:00 for the subsequent midnight-to-midnight period. Upon physical connection time t_{arr} , the charger acquires this updated price array alongside the user's specific departure time, t_{dep} , and total required energy, E_{req} .

We implement a fast, heuristic sorting algorithm suitable for basic microcontroller hardware. The controller first calculates the ideal charging duration, H_{req} , assuming the EV charges

continuously at the maximum hardware limit, P_{max} (where $H_{req} = E_{req}/P_{max}$). Next, the algorithm isolates the price array $C(t)$ exclusively within the user's connection window $[t_{arr}, t_{dep}]$. It sorts these available time slots in ascending order of cost and allocates active charging to the H_{req} cheapest hours. This scheduling logic generates a preliminary power setpoint, $P_{sch}(t)$. The algorithm assigns $P_{sch}(t) = P_{max}$ during the selected lowest-cost hours and $P_{sch}(t) = 0$ during all other periods. However, the equitable droop controller (detailed in Section 2.1) operates simultaneously and may curtail the actual power output to support grid voltage. If this happens, the EV will not reach E_{req} within the initially allocated H_{req} hours. To accommodate this physical constraint, the algorithm continuously tracks the delivered energy. If an energy deficit remains after the scheduled blocks conclude, the controller extends the charging session. Instead of searching for the next lowest-cost hour, the EV continues charging consecutively right after the scheduled period until it fulfills E_{req} . This continuous extension prioritizes rapid session completion over strict secondary price optimization, given that the residual energy requirement is typically small.

2.3 Adaptive V-P Control to Synthesize Grid and Cost Constraints

The local controller dynamically merges the economic schedule and the physical grid constraints to determine the final charging power, $P_{charge,i}(t)$. While the scheduling algorithm (Section 2.2) defines the operational windows, the equitable voltage droop (Section 2.1) enforces a physical boundary. To resolve these concurrent inputs, the controller calculates the instantaneous charging power by taking the minimum value between the scheduled economic setpoint and the local droop limit:

$$P_{charge,i}(t) = \min(P_{sch}(t), P_{droop,i}(V_i))$$

Under healthy grid conditions ($V_i \geq V_{th,i}$), the droop limit remains at P_{max} . This state allows the EV to execute its cost-optimized schedule without curtailment. Instead, during local undervoltage events ($V_i \leq V_{th,i}$), the droop controller actively reduces the allowable charging power. In this scenario, the grid support function overrides the economic schedule, forcing the charger to reduce its output.

The complete adaptive V-P strategy is represented in Fig. 1.

3 Simulation Framework

To validate the proposed control strategy, we conduct stochastic Monte Carlo simulations over continuous one-week horizons at a ten-minute temporal resolution. The environment relies on the Python-based Pandapower library to execute three-phase unbalanced power flow analyses. To manage the high computational load, we deploy the simulation environment on the DTU high-performance computing (HPC) cluster [13].

This section outlines the complete simulation architecture. We first classify the parameters into invariant, parametric, and

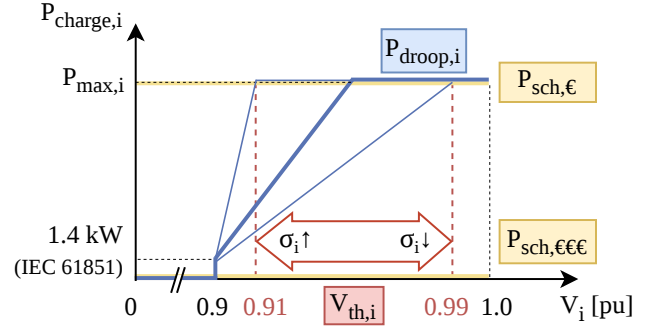


Fig. 1 Schematic representation of the proposed adaptive V-P charging strategy.

stochastic variables. Then, we describe the benchmark strategies, the operational scenarios, and define the key performance indicators used to quantify system behavior.

3.1 Invariant Framework Components

This subsection defines the static simulation parameters that remain constant across all Monte Carlo runs. We fix the physical network topology and the background residential load profiles to isolate the EV and the charging strategy impact.

3.1.1 Network Model: We evaluate the algorithms on a modified IEEE European LV asymmetric test feeder. To optimize computational efficiency, we use an electrically equivalent topology [14] with 116 buses, which preserves the original 55 residential load points. An 800 kVA distribution transformer connects the 11 kV medium-voltage primary network to the 0.416 kV LV secondary network.

3.1.2 Baseload Profiles: We model the background residential demand using the standardized 1-minute resolution load profiles provided within the original IEEE LV test feeder. To isolate the impact of EV charging strategies, we exclude alternative flexible loads and local distributed generation. Consequently, the simulation environment features no explicitly modelled heat pumps or residential photovoltaics.

3.2 Parametric Scenario Variables

This subsection describes the parameters that we change to define the different scenarios, but that are kept fixed within each scenario.

3.2.1 EV Penetration: We define EV penetration as the ratio between the number of EVs and households in the grid; each household can have a maximum of one EV. EV penetration can assume values between 0 and 100%, which is the case when 55 EVs are distributed across the 55 load nodes in the network.

3.2.2 Electricity Prices: In the simulations a composite electricity price signal is made available to the local charging controllers. The signal represents the mathematical sum of

dynamic day-ahead market prices and static DSO tariffs. The day-ahead component reflects the market clearing price at 1-hour intervals, which the market operator publishes daily at 13:00 for the following day. To emulate realistic network management, we integrate standard Danish seasonal DSO tariffs designed to incentivize overnight consumption. More specifically, DSO tariffs are designed in a three-tier structure: low price from 00:00 to 06:00, medium price from 06:00 to 17:00 and from 21:00 to 00:00, high price from 17:00 to 21:00.

While retail electricity bills include additional region- or provider-specific fixed charges, the day-ahead and DSO elements constitute the primary time-variable components available for optimization.

3.3 Stochastic Variables

To capture the unpredictability of residential EV utilization, we sample probability distributions derived from a TotalEnergies residential charging dataset for a French municipality. These distributions define both the EV charging technical characteristics and the EV charging sessions.

3.3.1 Spatial Allocation and Connection Times: At the initialization of a simulation, we randomly assign EV locations across the available residential nodes.

To model charging frequency accurately, we extract daily connection probabilities directly from the empirical dataset. For the vehicles that connect, we independently sample arrival and departure times.

3.3.2 Charging Power and Energy Demand: Aligning with the grid model, we simulate single-phase residential charging infrastructure. We assign each connected EV a maximum charging capacity of either 3.6 kW or 7.4 kW according to the dataset probabilities.

Because an EV's energy demand correlates with its hardware limits, we sample the required energy for each session from two distinct probability distributions linked to the assigned power.

3.4 Benchmark Charging Strategies

To evaluate the proposed adaptive control strategy, we consider three distinct benchmark strategies.

3.4.1 Uncontrolled Charging: The uncontrolled strategy represents standard plug-and-charge behavior without any grid or economic awareness. Upon arrival, the EV immediately charges at maximum power until it satisfies the required energy demand. We use this strategy to reflect the prevalent charging method in current distribution networks.

3.4.2 Traditional Voltage Droop: The traditional voltage droop strategy adds a grid-support element to the uncontrolled formulation. Unlike the proposed equitable strategy (which adapts parameters based on the electrical distance from the transformer), this control applies the same voltage activation threshold of $V_{th} = 0.95$ p.u. across all nodes. As the local voltage

drops below this threshold, the controller linearly curtails the active power from P_{max} to the IEC 61851 hardware minimum of 1.4 kW. If voltage decreases below 0.90 p.u., the controller stops charging. This strategy represents conventional local voltage regulation without any compensation for spatial fairness.

3.4.3 Traditional Price-Optimized Charging: The traditional price-optimized strategy employs the pure economic scheduling logic defined in Section 2.2 while completely ignoring network constraints. The controller calculates the required charging duration, sorts the available connection window by cost, and allocates maximum power to the cheapest periods. Because this strategy applies no voltage droop regulation, chargers across the network operate at full capacity simultaneously during cheap hours. We model this behavior to show the effects of grid-agnostic economic optimization.

3.5 Evaluation Metrics

We assess the performance of the different charging strategies using a hierarchical framework of three key performance indicators (KPIs): grid impact, spatial fairness, and cost savings.

3.5.1 Grid Impact: We consider grid impact as the primary performance metric. If a charging strategy violates physical network limits, we consider the grid service interrupted, making the other economic and fairness metric irrelevant. We evaluate grid impact primarily through node voltage deviations, aligning with the main objective of the proposed charging strategy. We define voltage compliance according to the EN 50160 power quality standard, which sets operational limits based on 10-minute intervals over a one-week period. Specifically, the local node voltage must stay within a $\pm 10\%$ range (0.9 to 1.1 p.u.) for at least 95% of the measured intervals. Additionally, the standard sets a hard lower limit, requiring that no 10-minute interval goes below 0.85 p.u. at any time. If in a simulation either of these constraints are violated, we classify it as failed.

3.5.2 Fairness: We evaluate operational fairness by analyzing the distribution of power curtailment across the network. Traditional strategies disproportionately curtail power to EVs located far from the main transformer. To quantify this spatial disparity, we classify the active chargers into three distinct spatial buckets based on their electrical distance: near, middle, and far. For each simulation run, we compute the average charging power curtailment within these zones. We then apply Jain's fairness index [15] to these averages, to obtain a single comparison metric. An index score approaching 1.0 indicates a fair distribution of curtailment in the network.

3.5.3 Charging Costs: To quantify the direct economic benefit to the end-user, we compute the final cost of the delivered energy. We calculate this metric as the average price paid per kWh across the charging sessions. Lower average unit costs indicate higher success in shifting the required energy demand

Table 1 Global Performance Assessment Across Scaling EV Penetrations

Charging Strategy	EV Penetration Level				
	20%	40%	60%	80%	100%
<i>Primary Constraint: Undervoltage Status (EN 50160)</i>					
Uncontrolled	Safe	Safe	Safe	Fail	Fail
Trad. ToU	Safe	Fail	Fail	Fail	Fail
Trad. Droop	Safe	Safe	Safe	Safe	Safe
Adaptive V-P	Safe	Safe	Safe	Safe	Safe
<i>Secondary Metric 1: Spatial Fairness (Jain's Index)</i>					
Uncontrolled	1.00	1.00	1.00	1.00*	1.00*
Trad. ToU	1.00	1.00*	1.00*	1.00*	1.00*
Trad. Droop	0.33**	0.38**	0.41	0.45	0.48
Adaptive V-P	0.72**	0.82**	0.87	0.88	0.88
<i>Secondary Metric 2: Average Charging Price (€/kWh)</i>					
Uncontrolled	0.18	0.18	0.18	0.18*	0.18*
Trad. ToU	0.11	0.11*	0.11*	0.11*	0.11*
Trad. Droop	0.18	0.18	0.18	0.18	0.18
Adaptive V-P	0.11	0.11	0.11	0.11	0.11

* These values are obtained during unviable operational states (EN 50160 voltage violation).

** Indicates negligible absolute curtailment ($< 0.5\%$), making the fairness index practically insignificant.

toward periods with low day-ahead distribution prices and distribution tariffs.

3.6 Scenarios

To evaluate the proposed charging control logic, we structure the simulation tests into two distinct phases.

3.6.1 Performance Assessment: To perform a broad comparison across the charging strategies, we evaluate them across a progressive EV penetration. We increase it in 20% increments, ranging from a 20% baseline up to 100%. For each penetration level, we perform 544 distinct Monte Carlo simulations, each with a one-week horizon to capture multi-day EV charging behavior. This aggregate assessment filters the strategies based on the primary grid impact constraints before evaluating their secondary economic and fairness metrics.

3.6.2 Detailed Scenario - 80% EV Penetration: Following the global assessment, we isolate an 80% EV penetration level for a detailed analysis. We select this specific scenario because it represents a realistic high-load condition where the network is sufficiently stressed to trigger the activation of the droop controllers. Within this scenario, we evaluate the spatial distribution of minimum voltages across all grid buses. To assess spatial fairness, we analyze how the competing algorithms distribute active power curtailments among the EV locations. Finally, we quantify the average charging price reduction

to verify the economic efficiency of the proposed strategy compared to the baselines.

4 Results and Discussion

4.1 Performance Assessment

Table 1 presents the aggregate performance of the evaluated charging strategies across increasing EV adoption levels. We evaluate node undervoltage as the primary operational constraint, following the EN 50160 standard as described in Sub-section 3.5.1. The results show that with uncontrolled charging the grid is compliant up to 60% penetration. In contrast, the traditional price-optimized strategy causes the earliest violations at 40% penetration due to charging synchronization. Both the traditional voltage droop and the proposed adaptive V-P strategies maintain compliance across all simulated penetration levels.

We report spatial fairness and average charging prices for all scenarios, flagging with an asterisk the metrics that correspond to unfeasible grid states. Uncontrolled and traditional price-optimized strategies achieve theoretical maximum fairness because they never curtail charging power. With traditional voltage droop, spatial fairness slightly increases at higher penetrations, since the lower voltages in the grid activate the droop controller also in EVs closer to the source. The proposed adaptive V-P strategy achieves significantly higher fairness

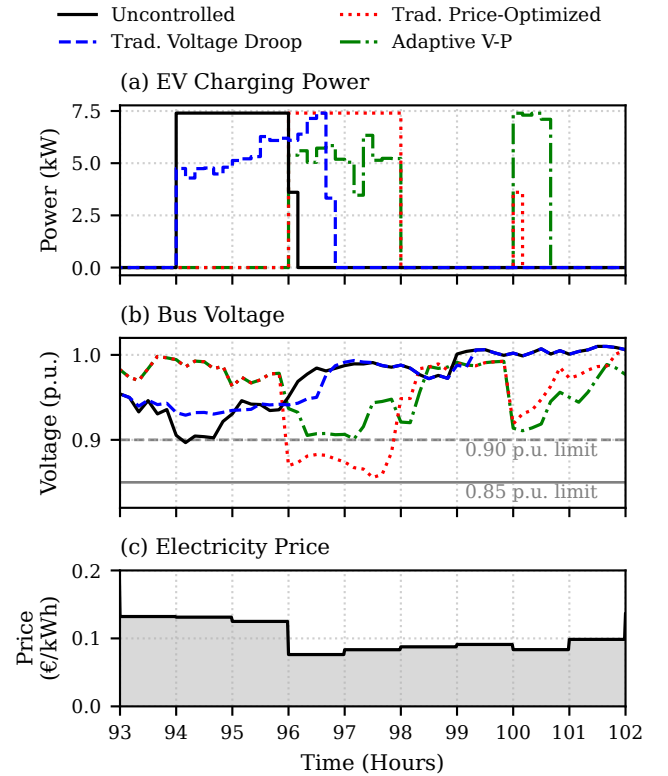


Fig. 2 Impact of EV charging strategies on the same bus. (a) EV Power, (b) Voltage, and (c) Electricity Price.

than the traditional voltage droop, since the EVs in the grid are more uniformly curtailed. From an economic perspective, the non-price-aware strategies feature average charging prices almost double those of the price optimized and adaptive V-P strategies.

4.2 Detailed Scenario

Following the global assessment, we isolate the 80% EV penetration scenario for an in-depth analysis.

4.2.1 Grid Impact: Fig. 2 shows the operation of the four charging strategies at a weak bus in the 80% penetration scenario. The uncontrolled strategy charges at the maximum power immediately upon arrival, causing the voltage to reach the 0.9 p.u. limit. The traditional droop controller activates at the same time, but actively adjusts charging power to keep the node voltage above the 0.9 p.u. level. Instead, the traditional price-optimized strategy delays charging until low-tariff hours. Since other EVs on the feeder perform a similar optimization, the grid experiences a voltage drop which is worse than with the uncontrolled strategy, reaching below the EN 50160 0.85 p.u. hard limit. The proposed adaptive strategy resolves the consequences of the synchronization. It still shifts demand to economical hours, but also applies active curtailment to stay within the 0.9 p.u. limit. To compensate for this curtailed power, both grid-aware strategies extend the total charging duration compared to the unconstrained ones.

To evaluate the network stress with the different strategies, Fig. 3 presents the distribution of the minimum voltages recorded at each grid nodes across all Monte Carlo runs in this scenario.

The traditional price-optimized strategy induces the lowest voltages, bringing the majority of buses below the 0.85 p.u. hard limit. Uncontrolled charging similarly drives the majority of buses below 0.85 p.u., though mainly with outlier values; most frequently bus voltages stay above this value but still below the 0.9 limit. The traditional voltage droop controller is the best performing from the voltage perspective, with no recorded minimum voltage below the 0.9 p.u. limit. Finally, the adaptive V-P strategy presents some far buses with voltages below 0.9 p.u. However, these transient deviations remain within the 5% temporal limit of the EN 50160 standard, and this strategy also ensures that the voltages are always above the 0.85 p.u. threshold.

4.2.2 Fairness: Fig. 4 shows the average power curtailment across the spatial buckets defined in 3.5.2, together with the corresponding Jain's fairness index. The uncontrolled and traditional price-optimized strategies apply zero active power curtailment, reaching a theoretical maximum Jain's index of 1.0. On the contrary, the traditional voltage droop disproportionately curtails EVs at buses distant from the transformer, resulting in a low fairness score of 0.45. Even though price-synchronization exposes the proposed adaptive V-P strategy to more critical network stress, the controller distributes active power curtailment more equitably. By requiring near buses to

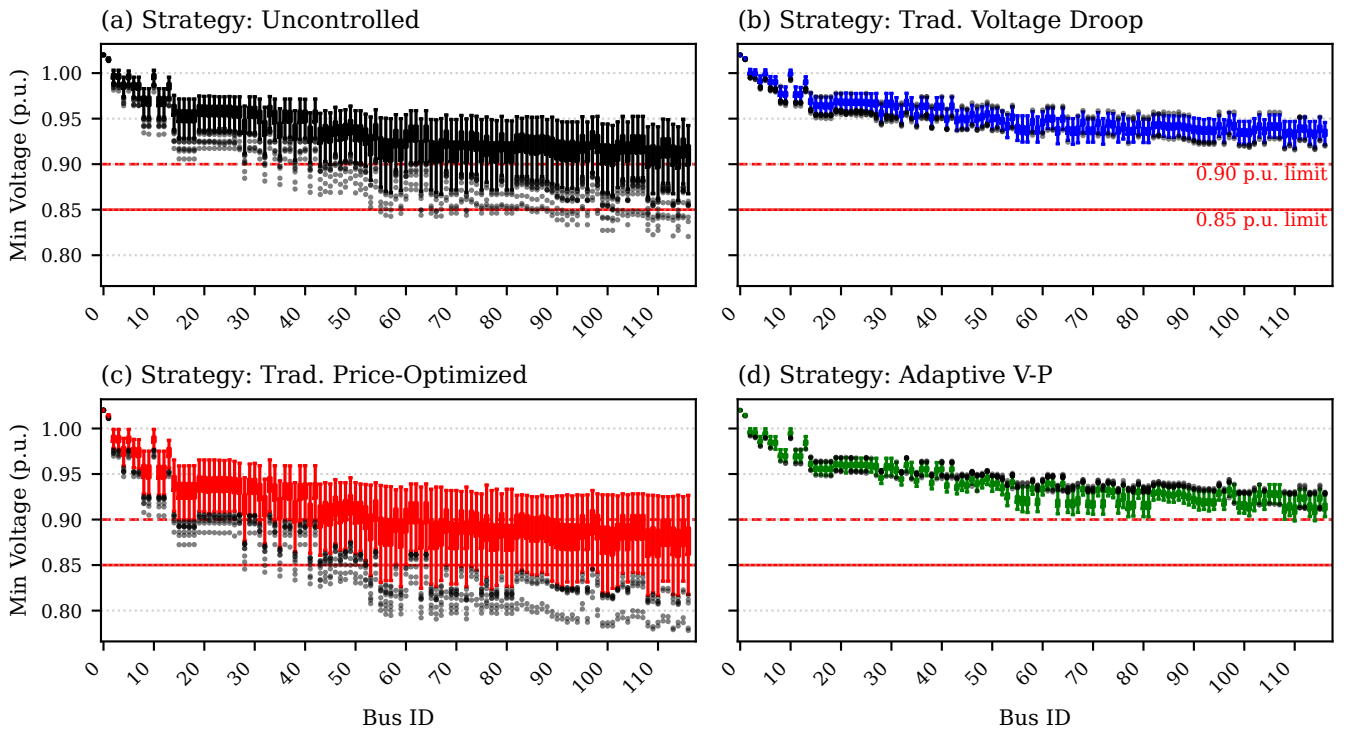


Fig. 3 Distribution of minimum voltage per bus across all Monte Carlo runs in the 80% EV penetration scenario. (a) Uncontrolled, (b) Traditional Voltage Droop, (c) Traditional Price-Optimized, (d) Adaptive V-P.

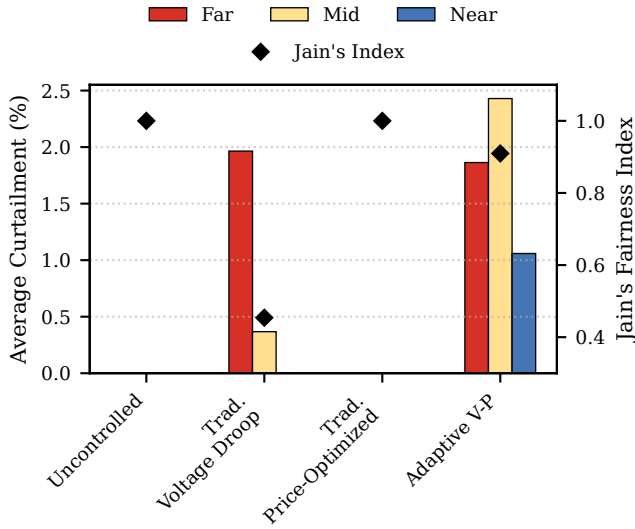


Fig. 4 Spatial fairness of EV charging strategies. Grouped bars show the average curtailment ratio (%) at Far, Mid, and Near locations from the substation. The secondary axis shows Jain's Fairness Index, where 1.0 represents perfect equality. Note: Uncontrolled and Trad. Price-Optimized strategies do not apply power curtailment, therefore their curtailment is 0%.

participate to the curtailment, it mitigates synchronized voltage drops without disadvantaging far grid nodes.

4.2.3 Cost Savings: Fig. 5 presents the distribution of the average charging price per kWh across all EVs during the Monte Carlo simulations. The uncontrolled and traditional voltage droop strategies operate blindly to market tariffs, resulting in median charging costs nearly double those of the price-aware alternatives. Moreover, the statistical distributions shows that the absolute minimum average price incurred by the non-price-aware controllers aligns closely with the maximum average price paid with the optimized strategies. Both the traditional price-optimized and proposed adaptive V-P strategies successfully minimize end-user costs, with the adaptive controller simultaneously complying with grid voltage limits.

4.2.4 Overall Performance Evaluation: Fig. 6 synthesizes the operational trade-offs of the evaluated charging strategies across the three KPIs within the 80% penetration scenario. The uncontrolled baseline achieves maximum theoretical fairness but incurs relatively high average charging costs and causes a voltage violation in 1.4% of the runs. The traditional price-optimized strategy minimizes user costs but sacrifices grid stability, bringing the voltage violation rate to 19.5%. The traditional voltage droop controller eliminates voltage violations but performs poorly with respect to spatial fairness (0.45) while maintaining high user costs (0.18 €/kWh). The proposed adaptive V-P strategy successfully resolves these competing operational objectives without compromising individual metrics. It keeps voltage compliance (0.0% violations) and grants

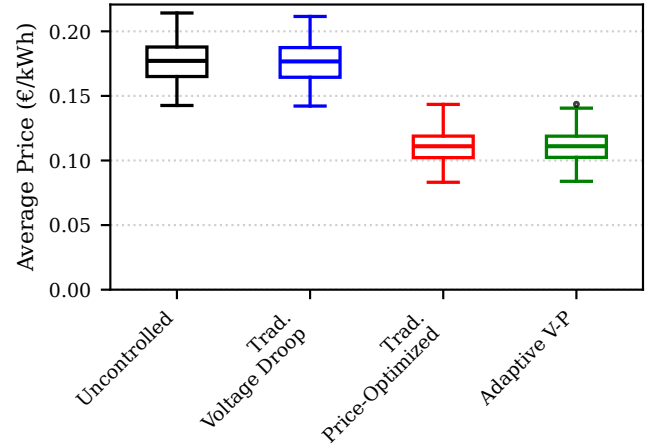


Fig. 5 Distribution of the average price paid per kWh by EVs across the different control strategies in the 80% penetration scenario.

the minimum average charging cost (0.11 €/kWh) alongside an equitable power curtailment distribution (0.88 Jain's Index).

5 Conclusion

In this paper we presented a decentralized EV charging strategy that balances spatial equity, grid compliance, and end-user price optimization while considering realistic data exchange

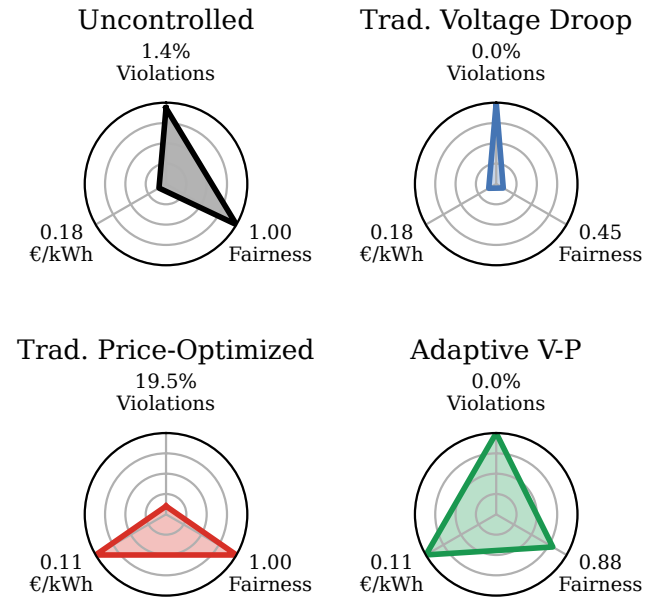


Fig. 6 Radar charts comparing the performance of the four evaluated charging strategies across three competing objectives: EN 50160 voltage violation probability, spatial fairness (Jain's Index), and average charging cost. The Adaptive V-P strategy demonstrates the most balanced compromise across all three KPIs compared to the other strategies.

in a low voltage grid. We validated the proposed strategy on the IEEE LV test feeder using empirical EV charging data and Monte Carlo simulations. We executed over ten years of aggregated simulation time at a ten-minute resolution to capture real-world stochastic behaviors. We benchmarked the adaptive V-P strategy against uncontrolled demand, a traditional voltage droop controller, and a traditional price-optimized algorithm. We evaluated these approaches using EN 50160 standard voltage compliance, spatial curtailment fairness, and average charging costs.

Results show that traditional price-optimized charging induces the most severe grid violations failing from the 40% penetration scenario, even though it successfully minimizes charging costs. Uncontrolled charging delays system failure until 80% penetration but doubles the average charging cost compared to price-aware alternatives. The traditional voltage droop respects the grid limits across all tested penetrations, but performs poorly in terms of fairness, disproportionately curtailing buses far from the transformer. Being blind to prices, it also performs poorly with respect to charging costs. In contrast, the proposed V-P strategy satisfies EN 50160 voltage constraints across all adoption levels. At the same time, it achieves average charging prices similar to traditional price-optimized charging, and manages to distribute power curtailment more evenly to sustain the grid during price-driven demand peaks.

Future research will expand this framework to incorporate localized photovoltaic generation, residential heat pump demand, and bidirectional charging. We will also validate the control architecture's scalability across other low-voltage network topologies and under alternative grid tariff structures.

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